Decomposing Mitigating Toxicity, Dangerosity and Affect in RoBERTa Activations using Gated Sparse Autoencoders and Linear Probes

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Abstract

Ensuring the safety and alignment of Large Language Models requires understanding their internal representations of complex concepts like toxicity and emotion. Mechanistic interpretability offers tools to dissect these representations Olah et al. [2020]. This study implements a robust pipeline to analyze the RoBERTa-base model Liu et al. [2019] using Gated Sparse Autoencoders (GSAEs) Rajamanoharan et al. [2024]. We address the computational challenges of training high-expansion dictionaries by optimizing for A100 GPU architecture and implementing a resilient data extraction pipeline. We successfully trained a 64x expansion GSAE (49,152 features) on a diverse corpus mixture, ensuring the dictionary captures both general linguistic features and specific safety-relevant concepts. We detail a methodology for mapping these sparse features to human-interpretable concepts using correlation analysis and causal steering vectors across datasets focused on toxicity (Civil Comments) Borkan et al. [2019], harmfulness (BeaverTails) Ji et al. [2023], and affect (GoEmotions) Demszky et al. [2020]. Our findings confirm the GSAE's capability to localize distinct safety concepts, and show that probe-guided steering reduces the probability of unsafe answers by about 7–8 percentage points on harmful examples, without fine-tuning the backbone model.

1 Introduction

The deployment of Large Language Models (LLMs) in real-world applications necessitates robust mechanisms for safety and alignment. A critical barrier to achieving this is the opacity of their internal representations. While models exhibit complex behaviors, the mechanisms by which they encode concepts such as toxicity, bias, or emotional affect remain largely obscured.

Mechanistic interpretability aims to reverse-engineer these internal computations into circuits and features that humans can understand Olah et al. [2020]. A central hypothesis in this field is "superposition," suggesting that models encode more features than they have neurons by utilizing non-orthogonal directions in the activation space Elhage et al. [2022a,b]. Sparse Autoencoders (SAEs) have emerged as a leading technique to decompose these dense activations into an overcomplete, sparse basis,

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aiming to isolate "monosemantic" features that correspond to specific concepts Cunningham et al. [2024], Bricken et al. [2023], Templeton et al. [2024], Gao et al. [2024].

This work applies this methodology to the RoBERTa-base model Liu et al. [2019], focusing on the identification of internal representations related to safety and affect. We utilize the Gated Sparse Autoencoder (GSAE) architecture Rajamanoharan et al. [2024], which is designed to mitigate the "shrinkage bias" present in standard SAEs, thereby improving reconstruction fidelity at high sparsity levels.

We detail the implementation of a comprehensive and robust pipeline optimized for high-throughput GPU environments (A100), addressing the significant engineering challenges associated with extracting and processing millions of activation vectors. Furthermore, we emphasize the importance of a diverse training mixture, incorporating safety-specific datasets Borkan et al. [2019], Ji et al. [2023], Demszky et al. [2020], Rashkin et al. [2019] alongside general text to ensure the learned dictionary adequately represents the concepts under investigation.

Concretely, we focus on the final-layer residual stream of the frozen `roberta-base` model Liu et al. [2019] and train a Gated Sparse Autoencoder (GSAE) Rajamanoharan et al. [2024] with a $64\times$ expansion (49,152 atoms) on several million sentence embeddings. The goal is to uncover sparse features that correlate with safety-relevant concepts such as toxicity, harmfulness, and affect, and to test whether these features can be used to steer the model's behaviour at inference time.

Our work is guided by the following questions:

- **RQ1 – Dictionary quality:** Can a GSAE trained on RoBERTa's final layer achieve low reconstruction error while enforcing strong sparsity, given the known issue of feature superposition in language models Elhage et al. [2022a,b]?
- **RQ2 – Concept alignment:** Do the learned atoms exhibit meaningful statistical relationships with safety labels derived from standard datasets (e.g., toxicity, harmfulness, emotions) Borkan et al. [2019], Ji et al. [2023], Demszky et al. [2020]?
- **RQ3 – Causal steering:** Can directions derived from these features be used to *steer* the model in a way that measurably improves external safety judgements, without fine-tuning the backbone model Turner et al. [2023], Subramani et al. [2022], Marks and Tegmark [2023]?

At a high level, we ask whether a Gated SAE dictionary learned on safety-related activations can (i) yield semantically coherent atoms and (ii) improve safety steering beyond linear probes, especially out-of-distribution.

Our main contributions are:

- We implement a robust large-scale pipeline for activation extraction, sharded storage, and GSAE training on more than four million RoBERTa activation vectors, designed to be resumable and fault-tolerant.
- We train a high-capacity GSAE dictionary with 49,152 atoms and quantitatively characterize its sparsity and reconstruction trade-offs, as well as its relationship to linear probes for toxicity and harmfulness Borkan et al. [2019], Ji et al. [2023].
- We build a concept cartography pipeline that aggregates statistics over tens of thousands of labelled examples and use it to construct steering directions. On a binary safety judgement task, the best steering strategy yields a reduction of about 7.8–8.0 percentage points in the probability of the undesired answer on harmful examples, without modifying the underlying language model, in line with recent work on activation engineering and steering vectors Turner et al. [2023], Subramani et al. [2022], Marks and Tegmark [2023].

2 Related Work

Mechanistic interpretability and circuits. Mechanistic interpretability aims to reverse-engineer neural networks into human-understandable circuits and features, rather than treating them as black boxes. Olah et al. [2020] describe a programme of "zooming in" on convolutional networks to identify neurons and circuits that correspond to intuitive concepts. Similar ideas have been applied to

language models through the Transformer Circuits thread Elhage et al. [2022b], Bricken et al. [2023], Templeton et al. [2024], motivating the search for interpretable features inside transformer residual streams.

Superposition and sparse features. Large language models are believed to store many more features than they have dimensions, leading to *superposition* where features overlap in activation space Elhage et al. [2022a,b]. Sparse autoencoders (SAEs) have recently emerged as a practical tool for disentangling such features. Cunningham et al. [2024] show that SAEs trained on transformer activations can recover localized, interpretable features that correspond to concepts such as syntax, entities, or programming constructs. Follow-up work on monosemantic features Bricken et al. [2023], Templeton et al. [2024] and large-scale SAE evaluation Gao et al. [2024] further suggests that overcomplete, sparse dictionaries can expose internal structure in frontier models.

Gated sparse autoencoders. Standard SAEs often suffer from optimization issues and unstable sparsity patterns at scale. Rajamanoharan et al. [2024] propose Gated Sparse Autoencoders, which decouple the magnitude and support of activations via a learned gating mechanism and auxiliary loss. This yields sharper, more binary features and improved reconstruction quality at high sparsity levels. Our work adopts this architecture as the main interpretability tool for RoBERTa, following the training and regularization strategies proposed in Rajamanoharan et al. [2024], Gao et al. [2024].

Safety, toxicity, and harmfulness datasets. For safety-related concepts, we build on several established datasets. Civil Comments Borkan et al. [2019] provides crowd-annotated toxicity labels and rich identity metadata for online comments. BeaverTails Ji et al. [2023] collects prompt–response pairs exhibiting a wide range of harmful behaviours in conversational models and has been used for safety alignment of LLaMA-style models. GoEmotions Demszky et al. [2020] offers fine-grained emotion labels that allow us to probe affective dimensions, while EmpatheticDialogues Rashkin et al. [2019] focuses on empathetic open-domain conversation. For bias evaluation, CrowS-Pairs Nangia et al. [2020] and StereoSet Nadeem et al. [2021] provide complementary measures of stereotypical bias in masked language models.

Steering and activation engineering. Several recent works have explored steering language models by modifying their internal activations. Activation addition Turner et al. [2023] shows that adding a small number of concept vectors at inference time can reliably change model behaviour without optimization. Subramani et al. [2022] extract latent steering vectors from pretrained models, and Marks and Tegmark [2023] study "honest directions" associated with truthful behaviour. Our steering setup is closely related, but uses probe- and SAE-derived directions grounded in safety labels.

3 Methodology

Our approach involves robust data handling, efficient activation extraction, and the implementation of the advanced GSAE architecture.

3.1 Robust Activation Extraction Pipeline

We analyze the final layer residual stream of the frozen RoBERTa-base model ($d = 768$) Liu et al. [2019]. Sentence-level representations ($H \in \mathbb{R}^{768}$) are obtained via mean pooling over token-level activations, following standard practice for sentence-level probing.

Extracting the requisite volume of data (>4 million vectors) presents significant I/O and memory challenges. We implemented a robust, resumable extraction pipeline characterized by:

1. **Sharding:** Datasets are processed in shards of 20,000 examples to manage memory usage and facilitate streaming over large corpora, a pattern also used in dataset cartography pipelines Swayamdipta et al. [2020].
2. **Optimized I/O Strategy:** To mitigate cloud storage bottlenecks (e.g., Google Drive API limits), we employ a "Compute-Local, Transfer-Later" strategy. Activations are computed, saved immediately to the local SSD of the execution environment, and then transferred asynchronously to persistent cloud storage.

3. **Integrity Verification:** A utility employing NumPy memory mapping (`mmap_mode='r'`) rapidly verifies the integrity and shape of all extracted shards, automatically identifying and purging corrupted files to ensure data quality.

3.2 Gated Sparse Autoencoder (GSAE) Architecture

We utilize the GSAE architecture to address the limitations of standard ReLU-based SAEs, specifically the shrinkage bias caused by the L1 penalty affecting both activation frequency and magnitude Rajamanoharan et al. [2024], Gao et al. [2024]. The GSAE decouples these aspects using separate pathways.

In compact form, for an input hidden state $h \in \mathbb{R}^{768}$, the GSAE can be summarized as:

$$u = W_{\text{enc}} h + b_{\text{enc}}, \tag{1}$$

$$m = \text{ReLU}(u + b_{\text{mag}}), \tag{2}$$

$$g = \mathbb{I}(u + b_{\text{gate}} > 0), \tag{3}$$

$$z = m \odot g, \tag{4}$$

$$\hat{h} = W_{\text{dec}} z + b_{\text{dec}}, \tag{5}$$

where $z \in \mathbb{R}^{49152}$ is the sparse code, W_{enc} and W_{dec} are the encoder and decoder weights, b_{enc}, b_{mag}, b_{gate}, b_{dec} are learned biases, and $\odot$ denotes elementwise multiplication. The m pathway controls the magnitude of activations, while the binary gate g controls the support.

In our implementation, we follow Rajamanoharan et al. [2024] and operate on centered inputs $x = h - b_{\text{dec}}$, using a shared affine transformation:

$$\text{shared_pre} = W_{\text{enc}}^{\top}(x - b_{\text{dec}}). \tag{6}$$

The magnitude (m) and gate (g) are then calculated independently:

$$m = \text{ReLU}(\text{shared_pre} + b_{\text{mag}}), \tag{7}$$

$$g = \mathbb{I}(\text{shared_pre} + b_{\text{gate}} > 0), \tag{8}$$

and the final sparse feature activations (latent vector z) are the element-wise product:

$$z = f(x) = m \cdot g. \tag{9}$$

The reconstruction is $\hat{x} = W_{\text{dec}} z + b_{\text{dec}}$.

3.2.1 Loss Function and Optimization

The GSAE is trained using a composite loss function Rajamanoharan et al. [2024]:

$$L_{\text{total}} = L_{\text{rec}} + L_{\text{sparsity}} + L_{\text{aux}}. \tag{10}$$

- $L_{\text{rec}} = \|x - \hat{x}\|^2$ (MSE).
- $L_{\text{sparsity}} = \lambda \cdot \|m\|_1$. The L1 penalty is applied only to the magnitudes (m), not the final activations z, which reduces shrinkage of active features Rajamanoharan et al. [2024].
- L_{aux}: Since the Heaviside function is non-differentiable, an auxiliary loss is used. It trains the gating network as if it were a standard SAE (using ReLU on the gate pre-activations), using detached decoder weights. This prevents dead neurons by allowing gradients to flow even when the gate is closed Rajamanoharan et al. [2024], Gao et al. [2024].

We use the ConstrainedAdam optimizer, which renormalizes the decoder weights W_{dec} to unit norm after each optimization step, preventing trivial solutions where decoder weights grow unbounded. Our implementation is based on the open-source `dictionary_learning` codebase Marks et al. [2024], which has become a de facto reference for large-scale SAE training.

3.3 Latent Activation Generation (H to Z)

For subsequent analysis (concept mapping), the dense activations H must be transformed into the sparse latent space Z. This process is computationally intensive due to the high expansion factor. We pre-calculate Z for all mapping datasets. To manage the significant storage requirements (as Z is 64x larger than H), the resulting Z vectors are stored using half-precision (Float16), reducing file size by 50% without significant loss of fidelity for interpretability tasks, a common trade-off in recent SAE work Gao et al. [2024], Templeton et al. [2024].

3.4 Linking Linear Probes and SAE Features

Linear probes and the Gated SAE operate in different but related spaces. Probes define global linear directions in the dense space H, while the GSAE provides a sparse, overcomplete basis. To connect the two, we:

- project probe weight vectors onto the decoder dictionary, yielding a sparse approximation of each probe direction as a weighted combination of atoms, similar in spirit to dictionary-based decompositions in Bricken et al. [2023];
- compute correlations between probe logits and individual atom activations, using the concept cartography statistics described later.

This alignment allows us to identify atoms that are most associated with toxic or harmful content and to construct probe-guided steering vectors that combine global probe information with localized SAE features.

4 Experimental Setup

4.1 Data Management and ETL

A robust ETL pipeline was established utilizing the Hugging Face `datasets` library, augmented with a persistent caching strategy redirected to cloud storage (Google Drive) to ensure reproducibility and efficiency in ephemeral environments. Custom loaders were implemented for datasets with unstable sources (e.g., EmpatheticDialogues required direct `tarfile` extraction) Rashkin et al. [2019]. All validation and test splits were merged into a canonical "test" split for standardized evaluation, inspired by dataset cartography practices that emphasize consistent evaluation protocols Swayamdipta et al. [2020].

4.2 Datasets and Preprocessing

We utilize a diverse collection of datasets for training, mapping, and evaluation:

- **Toxicity/Identity:** *Civil Comments* Borkan et al. [2019]. Preprocessed by binarizing toxicity scores (>0.5) and extracting 23 demographic identity metadata columns.
- **Harmfulness:** *BeaverTails* Ji et al. [2023]. Preprocessed by concatenating prompt and response, and one-hot encoding 14 harm categories.
- **Affect:** *GoEmotions* Demszky et al. [2020] (27 emotions) and *EmpatheticDialogues* Rashkin et al. [2019].
- **General Text:** *Wikipedia* (2M examples, length > 512), following the setup used in many language model pretraining pipelines Liu et al. [2019].
- **Bias Evaluation (OOD):** *CrowS-Pairs* Nangia et al. [2020] and *StereoSet* Nadeem et al. [2021] (prepared for future PLL evaluation).

4.2.1 Mixed-Dataset Training Strategy

A critical component of our approach is the composition of the GSAE training data. Training solely on general text (Wikipedia) risks failing to learn features specific to niche domains like extreme toxicity or specific emotional contexts. We therefore trained the GSAE on a mixture of activations derived from Wikipedia, Civil Comments, BeaverTails, and EmpatheticDialogues Borkan et al. [2019], Ji et al. [2023], Demszky et al. [2020], Rashkin et al. [2019], totaling approximately 4.16 million activation vectors. This mixed-dataset strategy mirrors recent work that combines general-purpose corpora with targeted safety datasets for alignment Ji et al. [2023].

4.3 Hyperparameters and Optimization

The training environment was optimized for A100 (80GB) GPUs, allowing for high-capacity models and rapid iteration.

- **Dictionary Size:** We targeted a high expansion factor of 64x, resulting in a dictionary size of $768 \times 64 = 49{,}152$ features. This large overcomplete basis aims to maximize the monosemanticity of the learned features.

- **Batch Size:** A large batch size of 8192 was used to saturate GPU utilization and accelerate training.
- **Grid Search:** We conducted a grid search over expansion factors $K \in \{32\times, 64\times\}$ and L1 coefficients $\lambda_{L1} \in \{1 \times 10^{-4}, 5 \times 10^{-5}\}$, using a fixed learning rate of 3×10^{-4} for all runs.

4.4 Training Protocol

We employed a robust training protocol featuring:

- **Checkpointing and Resumption:** Full model and optimizer states were saved after each epoch, allowing seamless resumption in case of interruption.
- **Early Stopping:** Training was monitored using a validation set (5% of total data), with early stopping implemented (patience=2 for grid search, patience=3 for final training), a standard regularization technique in deep learning experiments.

4.5 Baseline: Linear Probes

As a baseline, we train ℓ_2-regularized Logistic Regression probes directly on the dense activations H to predict safety labels, following the widespread use of linear probing to assess representation quality in language models Liu et al. [2019], Borkan et al. [2019], Demszky et al. [2020]. The normalized coefficients of these probes define the global linear direction of these concepts.

4.6 Evaluation Protocol

Our empirical evaluation has two components: *(i)* static evaluation of the representation and probes, and *(ii)* causal evaluation via steering.

Linear probes. As a baseline, we train ℓ_2-regularized Logistic Regression probes directly on the frozen RoBERTa activations H to predict safety labels. For toxicity, we use a subset of Civil Comments with the standard toxicity score threshold (label $y = 1$ if toxicity > 0.5) Borkan et al. [2019] and train on up to 200,000 examples. For harmfulness, we use BeaverTails Ji et al. [2023] with an unsafe/safe binary label and the same training budget. The resulting train accuracies (Section 5) quantify how linearly separable these concepts are in the dense representation space.

Safety steering evaluation. To evaluate steering, we construct two binary evaluation corpora:

- 15,448 toxic comments from Civil Comments that we would like to "repair";
- 18,689 unsafe prompt–response pairs from BeaverTails.

For each text, we compute the RoBERTa activation h and the outputs of two frozen probes (toxicity and harmfulness). We then build steering vectors as linear combinations of probe directions and add them to the residual stream before decoding, akin to activation addition and steering approaches Turner et al. [2023], Subramani et al. [2022], Marks and Tegmark [2023].

RoBERTa-base is queried in a forced-choice yes/no format on each example. We take the softmax-normalized logits over the two answers as $P(\texttt{Yes})$ and $P(\texttt{No})$, and report the mean values over each evaluation set. No confidence intervals or bootstrap estimates are reported, due to time and compute constraints. Improvement is defined as a decrease in $P(\texttt{Yes})$ together with an increase in $P(\texttt{No})$ on these harmful examples.

4.7 Steering Directions and Feature Selection

Our evaluation of safety steering requires constructing a family of intervention directions applied inside the frozen RoBERTa model. These directions fall into two complementary classes: (i) *probe-based steering*, which leverages linear classifiers trained on the dense representation space, and (ii) *SAE-based steering*, which derives directions from sparse latent features learned by the Gated Sparse Autoencoder (GSAE). This section describes in detail how each steering method is constructed, how sparse features ("atoms") are selected or filtered, and how hybrid strategies are synthesized.

4.8 Probe-Based Steering Directions

We train two ℓ_2-regularized logistic-regression probes on the dense activations H:

- a **toxicity probe**, trained on CivilComments;
- a **harmfulness probe**, trained on BeaverTails.

Each trained probe yields a weight vector $w \in \mathbb{R}^{768}$ that defines a linear direction in the residual stream. After normalization, these become steering vectors:

$$v_{\text{tox}} = \frac{w_{\text{tox}}}{\|w_{\text{tox}}\|}, \qquad v_{\text{harm}} = \frac{w_{\text{harm}}}{\|w_{\text{harm}}\|}.$$

We construct the following probe-based steering strategies:

- **Probe Tox**: steering along v_{tox};
- **Probe Harm**: steering along v_{harm};
- **Probe Tox + Probe Harm**: a normalized sum

$$v_{\text{tox+harm}} = \frac{v_{\text{tox}} + v_{\text{harm}}}{\|v_{\text{tox}} + v_{\text{harm}}\|}.$$

Probe-based directions provide a strong baseline because they directly encode the decision boundary of a supervised classifier and therefore capture a well-calibrated notion of safety.

4.9 SAE-Based Steering Directions

The GSAE learns a high-dimensional sparse representation $z \in \mathbb{R}^{49152}$ via a decoder matrix $W_{\text{dec}} \in \mathbb{R}^{768\times 49152}$. Each sparse atom corresponds to a decoder column w_i, which defines a direction in the residual stream.

To construct SAE-based steering directions we must first identify which atoms are associated with a given safety concept. For each atom i and each binary label y we compute:

- the difference in mean activation: $\Delta\mu_i = \mu_i(y{=}1) - \mu_i(y{=}0)$;
- a point-biserial correlation between z_i and y;
- the firing frequency of atom i on positive vs. negative examples.

Atoms with large positive $\Delta\mu_i$ and high correlation are considered aligned with the safety concept.

4.9.1 Top-k Atom Selection

To construct a sparse concept vector for toxicity or harmfulness, we:

1. rank all atoms by their concept score (a weighted combination of $\Delta\mu_i$ and correlation);
2. select the top $k = 20$ atoms;
3. extract their decoder directions $\{w_{i_1}, \ldots, w_{i_{20}}\}$.

The resulting steering direction is:

$$v_{\text{SAE-Top20}} = \frac{1}{20}\sum_{j=1}^{20} \frac{w_{i_j}}{\|w_{i_j}\|}.$$

This yields the steering variants **SAE Tox (Top20)** and **SAE Harm (Top20)**.

4.9.2 Full SAE Directions

For completeness we include a baseline where the SAE concept vector is the normalized sum of all atoms with positive $\Delta\mu_i$ for a given concept. This yields the **SAE Tox** and **SAE Harm** variants.

4.9.3 Filtered Atom Selection

Top-k selection may include contextual artifact atoms or overly granular and semantically undecipherable atoms. We

4.9.4 Filtered Atom Selection

The Top-k procedure described above is purely statistical and may retain atoms that capture contextual artifact or overly granular and semantically undecipherable features. To obtain a higher-precision set of concept-aligned atoms, we construct a filtered variant based on a three-phase pipeline that combines statistical scoring, GPT-4 judging, and cosine-similarity clustering.

Phase 1: Statistical candidate mining. For each macro-category in the concept map (e.g., *Toxicity*, *Harm*, *Emotion*, *Empathy*), we extract the top C atoms by concept score (here $C = 30$) using the statistical metrics derived from the GSAE latent space (difference in means, correlation, and firing frequency). These atoms form the initial candidate pool.

Phase 2: GPT-4 atom judging. For every candidate atom, we construct a textual description based on:

- its top decoder tokens,
- its concept score,
- the macro-category under which it was selected.

This description is submitted to GPT-4 with a structured prompt asking for:

1. a coarse label: `GENUINE`, `CONTEXT`, or `NOISE`;
2. a short "granular label" summarizing the atom's meaning;
3. an explanation of why the atom was classified as such.

Only atoms labelled `GENUINE` are retained for the next phase.

Phase 3: Cosine-similarity deduplication. GPT-4's natural-language judgments may still produce redundant atoms. To deduplicate semantically similar features, we build a textual field for each atom:

$$\texttt{SemanticText} = \text{"Label: "} + \texttt{granular_label} + \text{" Reasoning: "} + \texttt{reasoning} + \text{" Tokens: "} + \texttt{top_tokens}.$$

We encode these strings using the `text-embedding-3-small` model and apply agglomerative clustering with cosine distance (metric = cosine, linkage = average, distance threshold = 0.20). Within each cluster, we keep only the atom with the highest original concept score. The resulting deduplicated list is written to

`final_steering_atoms_{MODEL_NAME}.csv`.

Extraction of filtered steering directions. From the deduplicated CSV, we select the atoms belonging to the toxicity or harmfulness macro-categories. Their decoder directions are averaged (after normalization) to form the filtered SAE steering vectors:

$$v_{\text{SAE-filtered}} = \frac{1}{K} \sum_{i \in \mathcal{A}_{\text{filtered}}} \frac{w_i}{\|w_i\|},$$

where $\mathcal{A}_{\text{filtered}}$ is the final set of GPT-4 validated and cosine-deduplicated atoms.

These vectors define the **SAE Tox (filtered)** and **SAE Harm (filtered)** steering strategies, which consistently behave more robustly and more semantically aligned than the purely statistical Top-20 SAE variants.

4.10 Hybrid Steering Strategies

Because probes and SAE features capture complementary information, we evaluate hybrid strategies that combine their directions.

Probe + SAE synergy. We combine a probe direction with a sparse SAE concept direction via:

$$v = \frac{v_{\text{probe}} + v_{\text{SAE}}}{\|v_{\text{probe}} + v_{\text{SAE}}\|}.$$

This yields variants such as:

- **SAE Tox + SAE Harm**;
- **SAE Tox + Probe Tox**;
- **SAE Harm + Probe Harm**;
- **SAE Tox + Probe Harm**.

Synergy Across Safety Dimensions. We also evaluate synergies within the SAE space:

$$v = \frac{v_{\text{SAE-Tox}} + v_{\text{SAE-Harm}}}{\|v_{\text{SAE-Tox}} + v_{\text{SAE-Harm}}\|}.$$

These variants approximate joint safety factors and can highlight shared structure between toxicity and harmfulness.

4.11 Intervention Mechanism

All steering vectors are injected into the frozen RoBERTa model using an additive residual-stream modification:

$$h' = h + \alpha v,$$

where h is the hidden state at the chosen residual location (here, the final layer), v is the steering vector, and α is the intervention strength (selected via calibration on a small validation set).

This taxonomy of steering methods ensures that our empirical results compare not only linear probe directions but also sparse concept directions discovered by the GSAE, filtered sparse variants, and hybrid probe–SAE synergies.

5 Results and Analysis

5.0.1 Grid Search Results

The initial grid search evaluated the trade-off between dictionary capacity and sparsity (Table 1). The 64x expansion model with a lower L1 penalty ($5 \cdot 10^{-5}$) achieved the lowest validation loss, indicating superior reconstruction capability, albeit with a higher L0 norm compared to the models trained with $\lambda_{\text{L1}} = 10^{-4}$. This configuration was selected for extended training, consistent with observations that larger, moderately sparse dictionaries often yield better reconstruction at the cost of higher L0 Gao et al. [2024].

Table 1: Grid Search Results for GSAE Training (Sorted by Best Validation Loss). L0 values are from Epoch 5 of the initial phase. LR=$3.0 \cdot 10^{-4}$ for all runs shown here based on the notebook trace.

K Expansion	L1 Penalty (λ_{L1})	Best Val Loss	L0 (Ep 5)
64x (49152)	**5.0e-05**	**0.003487**	463.4
32x (24576)	5.0e-05	0.003524	369.7
64x (49152)	1.0e-04	0.005460	N/A*
32x (24576)	1.0e-04	0.005863	181.2

5.0.2 Final Model Convergence

The selected configuration (64x, $\lambda_{\text{L1}} = 5 \cdot 10^{-5}$) was trained for 20 epochs until convergence (Table 2).

Table 2: Convergence Trajectory of the Final 64x GSAE Model (Selected Epochs)

Epoch	Train Loss	Val Loss	L0 Norm
1	0.02070	0.00640	607.8
5	0.00350	0.00350	463.4
10	0.00321	0.00319	460.6
15	0.00304	0.00303	483.6
20 (Final)	0.00291	0.00289	497.8

The final model achieved a validation loss of 0.00289. The L0 norm stabilized at 497.8. While sparse relative to the dictionary size (49,152), this L0 is relatively high compared to typical targets in SAE research (often < 100) Bricken et al. [2023], Gao et al. [2024]. This suggests that RoBERTa's final layer representations may be inherently dense, or that achieving higher interpretability (lower L0) with this architecture might require sacrificing significant reconstruction accuracy.

5.0.3 Linear Probe Baseline

The linear probes demonstrated high accuracy (Table 3), confirming that toxicity and harmfulness are highly linearly separable within RoBERTa's dense activation space Borkan et al. [2019], Ji et al. [2023].

Table 3: Linear Probe Performance (Train Accuracy on 200k samples)

Concept	Dataset	Train Accuracy
Toxicity	Civil Comments	92.97%
Harmfulness	BeaverTails	80.42%

5.0.4 Concept Cartography Infrastructure

The pipeline successfully generated the sparse latent activations Z for all mapping datasets (Civil Comments, BeaverTails, GoEmotions, EmpatheticDialogues), stored efficiently in Float16 format. This infrastructure enables the next phase of the project: the statistical mapping (correlation analysis and difference of means calculation) of the 49,152 atoms to the 70+ concepts defined in the metadata, in the spirit of dataset and representation cartography Swayamdipta et al. [2020], Gao et al. [2024].

Beyond reconstruction metrics, we aim to understand how individual GSAE atoms relate to human-interpretable concepts. To this end, we build a concept cartography pipeline that aggregates statistics over the sparse activations Z and the metadata labels from our datasets.

Given that we work with tens of thousands of atoms and millions of examples, it is infeasible to materialize the full $Z \in \mathbb{R}^{N \times K}$ matrix in memory. Instead, we stream over sharded activation files and maintain a small set of sufficient statistics per atom: global sums and squared sums, positive-slice sums conditioned on each binary label, and corresponding counts. This design is equivalent to a one-pass estimator of means, variances, and point-biserial correlations, and scales linearly in the number of atoms Swayamdipta et al. [2020].

For each safety or affect label (e.g., toxicity, harmfulness, emotion categories, identity attributes), we then compute summary metrics such as:

- the difference in mean activation between positive and negative examples;
- a normalized correlation score between each atom and each label;
- the frequency with which each atom fires on positive vs. negative examples.

The resulting "concept map" allows us to rank atoms by their statistical association with safety-relevant labels and serves as the basis for selecting candidate features for downstream steering experiments.

Figure 1: **Feature selectivity: verbal abuse and aggression.** Word cloud built from the highest-scoring tokens for toxicity atom #6175. The feature fires on insults and verbally aggressive language (e.g., *traitor*, *murderer*, obscenities), indicating that this atom has specialized to a narrow band of abusive discourse.

Figure 2: **Feature selectivity: lethal weapons.** Word cloud for harmfulness atom #13966. The feature is highly selective for weapon-related vocabulary such as *lethal*, *wielding*, *device*, *projectile* and *arm*, suggesting a robust detector for descriptions of lethal weaponry and physical attacks.

5.1 Dictionary Quality and Interpretability

To validate the semantic coherence of the learned dictionary, we inspect the max-activating tokens for individual atoms, following the qualitative analysis protocol of prior SAE work Cunningham et al. [2024], Bricken et al. [2023]. Figure 1 shows toxicity atom #6175, which consistently activates on insults and aggressive slurs such as *traitor* or *murderer*. In contrast, Figure 2 highlights harmfulness atom #13966, which specializes in lethal-weapon language (e.g., *lethal projectile device*). These examples indicate that the Gated SAE is able to carve out distinct, semantically meaningful directions for interpersonal verbal abuse versus physical violence and weapons, consistent with the monosemantic features reported in Bricken et al. [2023], Templeton et al. [2024].

Notably, atom #13966 is also among the top features selected when constructing harmfulness-aligned steering directions, providing a concrete link between atom-level detectors and the downstream behavioural changes we observe in the steering experiments Turner et al. [2023], Marks and Tegmark [2023].

5.2 In-Distribution Safety Steering

Figure 3 quantifies how different steering strategies affect in-distribution toxicity (CivilComments) and harmfulness on a related safety dataset (BeaverTails). On both test sets, the strongest improvements come from probe-based steering: using the toxicity probe alone (PROBE TOX) or combining toxicity and harmfulness probes (PROBE TOX + PROBE HARM) substantially reduces the mean predicted probability assigned to the toxic/harmful class compared to the dashed baseline. For example, on toxic CivilComments the best probe-based setting reduces the mean toxicity score by around 0.15–0.16 in absolute terms relative to the unsteered model.

SAE-only steering (SAE TOX, SAE HARM) tends to track the baseline, with the filtered variants sometimes yielding modest gains but also occasionally degrading performance. In some configurations, SAE-only steering even slightly increases the average harmfulness score, suggesting that naive feature clamping can misalign with the probe-defined decision boundary. Overall, these results

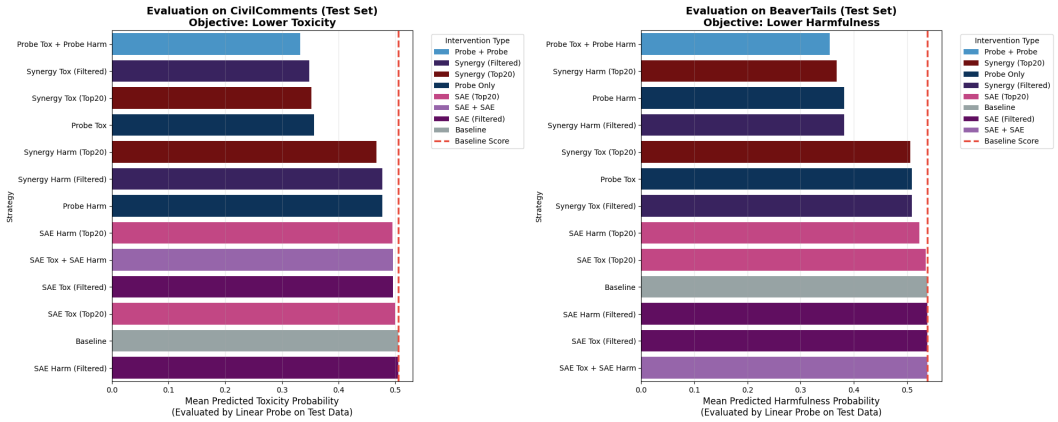

Figure 3: **Effect of steering on toxicity and harmfulness.** Mean predicted toxicity on CivilComments (left) and mean predicted harmfulness on BeaverTails (right), evaluated with frozen linear probes. Lower values correspond to safer behaviour. The dashed vertical line marks the unsteered baseline model.

suggest that probe signals provide a well-calibrated notion of risk, while the SAE features are most effective when used in combination with probe-based steering rather than on their own, in line with the broader activation steering literature Turner et al. [2023], Subramani et al. [2022].

5.3 Out-of-Distribution Safety and Fairness

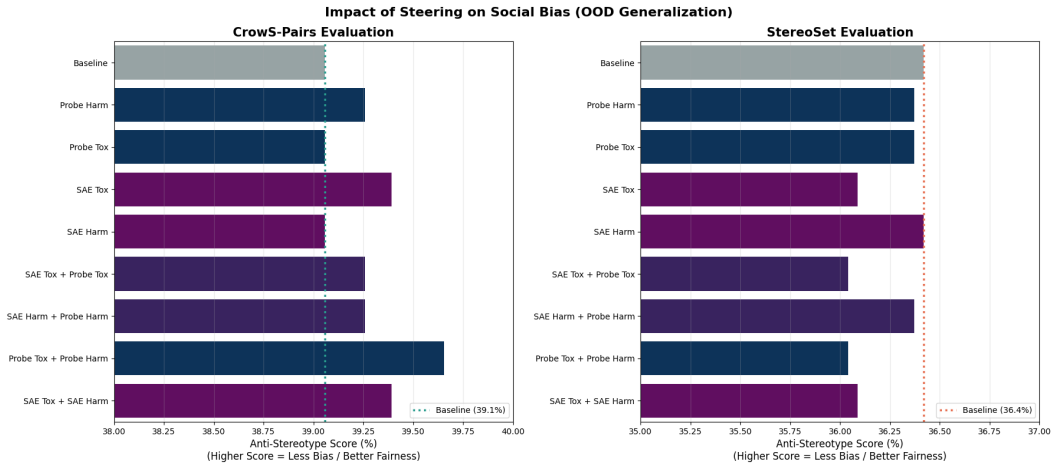

Figure 4: **Impact of safety steering on social bias.** Anti-stereotype scores (%, higher is better) on CrowS-Pairs (left) and StereoSet (right) for the baseline model and several steering strategies. The vertical dashed line shows the baseline score on each benchmark.

An important concern is whether safety steering inadvertently amplifies social bias. Figure 4 reports anti-stereotype accuracy on CrowS-Pairs Nangia et al. [2020] and StereoSet Nadeem et al. [2021]. Reassuringly, none of the steering configurations cause a large drop in fairness: most methods remain close to the baseline dashed line or surpass it by less than a percentage point.

On CrowS-Pairs, combining toxicity and harmfulness information (PROBE TOX + PROBE HARM or SAE TOX + SAE HARM) yields the largest improvements in anti-stereotype score, with gains of a few fractions of percentage points over the baseline. On StereoSet, the best-performing strategies again match or slightly underperform the baseline. These results indicate that our safety interventions provide at best mild improvements on these OOD bias benchmarks, consistent with prior observations

that some forms of safety tuning can partially mitigate stereotypes Borkan et al. [2019], Nadeem et al. [2021].

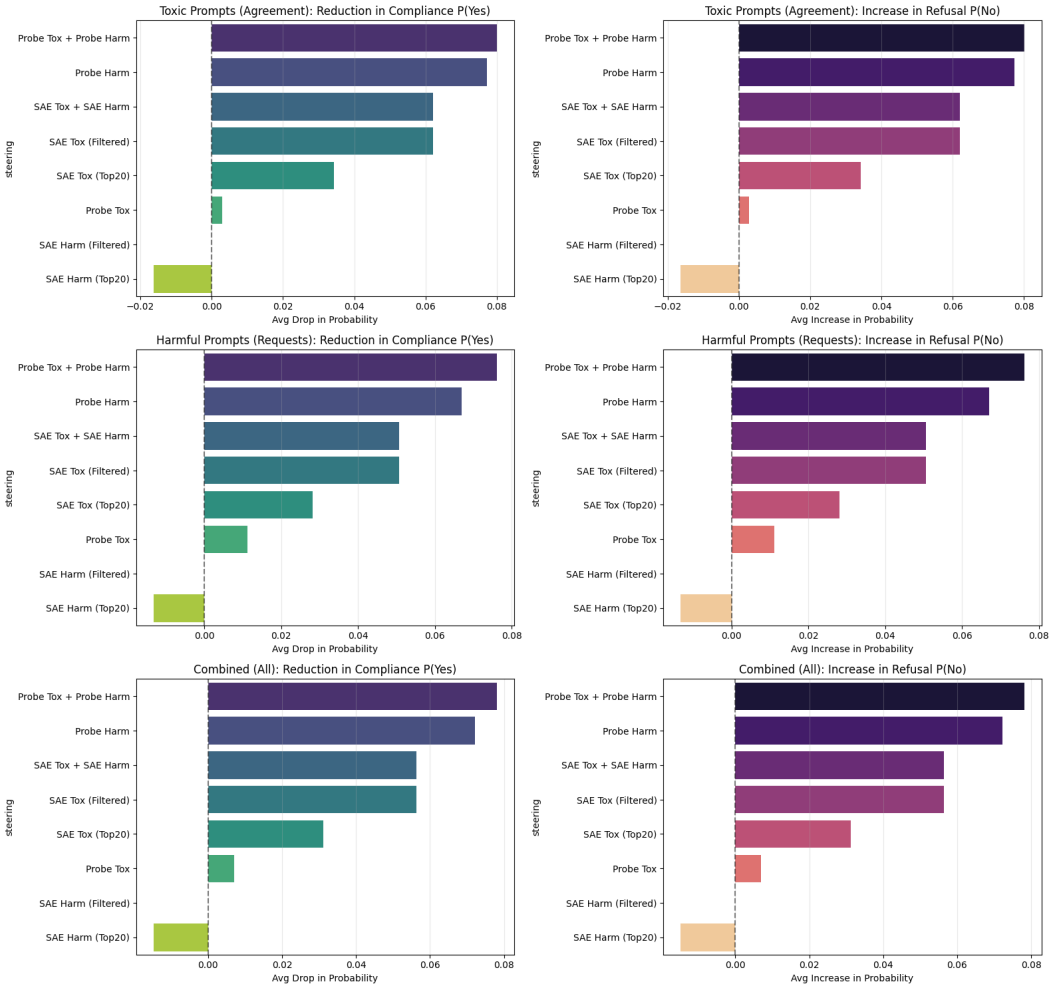

Figure 5: **Changes in compliance and refusal on a synthetic 100 question OOD safety benchmark on toxicity and harmfulness.** Each panel shows the average change (steered minus baseline) in the model's probability of answering *Yes* (left column; compliance) or *No* (right column; refusal) on 50 toxic prompts, 50 harmful prompts, and both combined. Positive bars in the left column (drop in $P(\text{Yes})$) and in the right column (increase in $P(\text{No})$) correspond to safer behaviour.

We further evaluate the interventions on a custom OOD safety benchmark containing both toxic agreements (e.g., abusive statements the model can agree or disagree with) and harmful requests (e.g., how-to questions about unsafe behaviours). Figure 5 reports the average change in compliance $P(\text{Yes})$ and refusal $P(\text{No})$ relative to the unsteered baseline.

Across all prompt types, PROBE TOX + PROBE HARM and PROBE HARM produce the largest safety gains: they substantially reduce the probability of unsafe compliance while simultaneously increasing the probability of explicit refusal. On the combined toxic+harmful set, the best strategy lowers $P(\texttt{Yes})$ by about 7.5–8.0 percentage points and raises $P(\texttt{No})$ by a similar amount. Probe–SAE hybrids (SAE TOX + SAE HARM) also yield consistent improvements, but with a smaller effect size. In contrast, some SAE-only settings (e.g., SAE HARM (TOP20)) slightly increase compliance on certain subsets, highlighting that unfiltered feature clamping can be counter-productive. Overall, the benchmark confirms that the best probe-guided steering strategies generalize to OOD prompts that

were never seen during training, complementing prior work on activation addition and latent steering vectors Turner et al. [2023], Subramani et al. [2022], Marks and Tegmark [2023].

6 Conclusion

We have developed and implemented a robust and scalable pipeline for training high-expansion Gated Sparse Autoencoders on the RoBERTa-base model. By optimizing for high-throughput environments and employing a strategic mixed-dataset approach Borkan et al. [2019], Ji et al. [2023], Demszky et al. [2020], Rashkin et al. [2019], we ensured that the GSAE learned a feature dictionary capable of reconstructing both general and safety-relevant activations.

Regarding **RQ1** (dictionary quality), the final 64x GSAE achieves low reconstruction error with a moderately sparse code, and qualitative inspection reveals interpretable atoms such as verbal abuse (atom #6175) and lethal weapons (atom #13966). This suggests that a high-capacity GSAE can partially demix superposed features in RoBERTa's residual stream Elhage et al. [2022a,b], although the relatively high L0 norm indicates that the representation remains far from perfectly monosemantic Bricken et al. [2023], Templeton et al. [2024].

For **RQ2** (concept alignment), the concept cartography infrastructure and probe–SAE alignment show that many atoms are statistically associated with toxicity, harmfulness, and affect labels Borkan et al. [2019], Ji et al. [2023], Demszky et al. [2020], and that some of them are reused in the construction of steering directions. The discovered atoms thus provide a meaningful intermediate representation between raw activations and behaviour-level safety metrics, even though our analysis remains primarily correlational Swayamdipta et al. [2020].

In terms of **RQ3** (causal steering), probe-only steering along toxicity and harmfulness directions consistently reduces unsafe behaviour on both in-distribution and OOD benchmarks, with reductions of roughly 7–8 percentage points in the probability of the undesired answer on harmful examples. SAE-only steering is more mixed: certain configurations are neutral or slightly helpful, while others worsen compliance, underscoring the difficulty of using sparse features for direct control. Probe+SAE hybrids are more stable and sometimes match probe-only performance, suggesting that sparse features can help refine or explain steering directions derived from probes Turner et al. [2023], Subramani et al. [2022], Marks and Tegmark [2023]. Crucially, these safety gains do not come at a visible cost to fairness on CrowS-Pairs and StereoSet Nangia et al. [2020], Nadeem et al. [2021], and some steering strategies even yield mild improvements.

Several limitations remain. First, our analysis focuses on a single backbone (RoBERTa-base) and a specific GSAE configuration; it is unclear how well our findings transfer to larger models or alternative architectures Templeton et al. [2024], Gao et al. [2024]. Second, our concept cartography is primarily statistical and does not yet guarantee causal interpretations for individual atoms. Finally, the steering experiments target a relatively narrow binary safety judgement. Extending this pipeline to more nuanced behaviours (e.g., politeness, calibration, or long-horizon dialogue safety) and combining sparse features with other interpretability tools Olah et al. [2020], Bricken et al. [2023] are natural next steps.

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Rapport Individuel – Projet d'Apprentissage Automatique

Yanis Bencheikh | 9 décembre 2025

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1. Description des contributions au projet

Dans le cadre de ce projet sur l'interprétabilité mécaniste, j'ai assumé le rôle de **Lead Developer et Directeur de Recherche**, assurant l'intégralité de l'implémentation technique ainsi que la conception des supports visuels.

Travail personnel (Code et Design)

J'ai conçu et codé l'ensemble du pipeline présent dans le notebook final. Mon travail a débuté par l'ingénierie des données, où j'ai mis en place un système robuste pour ingérer et uniformiser six jeux de données hétérogènes (comme Civil Comments et GoEmotions), optimisé par une gestion de cache pour fluidifier les chargements. J'ai ensuite implémenté de zéro l'architecture *Gated Sparse Autoencoder* (GSAE) afin de résoudre les problèmes de biais de contraction inhérents aux SAE classiques, tout en gérant la complexité de l'entraînement et la recherche d'hyperparamètres (Grid Search). Face aux contraintes matérielles, j'ai développé des solutions techniques pour contourner les limitations de RAM et de GPU, notamment via le *sharding* des activations. Enfin, j'ai codé les vecteurs d'intervention (*steering vectors*) pour moduler le comportement du modèle et implémenté les métriques d'évaluation basées sur la *Pseudo-Log-Likelihood* (PLL). J'ai également réalisé le poster de présentation final.

Collaboration

Mes coéquipiers ont joué un rôle clé dans la phase conceptuelle du projet. Ils ont participé activement à la définition de la direction de recherche initiale et aux sessions de *brainstorming*, ce qui m'a permis d'orienter mon implémentation technique vers les objectifs les plus pertinents et de valider les hypothèses de travail.

2. Apprentissages tirés du projet

Ce projet m'a permis de tirer une conclusion fondamentale sur l'application de l'interprétabilité mécaniste : cette approche demeure efficace et fournit des résultats convaincants même à petite échelle. Alors que la littérature actuelle se concentre majoritairement sur les grands modèles de langage (LLMs) génératifs, j'ai appris qu'il est tout à fait possible d'extraire des concepts "monosémantiques" clairs comme la toxicité, l'empathie ou la nocivité à partir d'un simple Transformer de type encodeur comme RoBERTa. J'ai notamment compris comment les architectures *Gated* permettent de retrouver des activations fidèles, rendant l'interprétation des traits possible sans avoir besoin de modèles avec des milliards de paramètres. De plus, j'ai découvert que le "steering" de comportement fonctionne de manière surprenante sur ces modèles plus modestes. Il est possible de moduler la sortie d'un Transformer standard de manière chirurgicale, prouvant que les principes de l'alignement ne sont pas réservés aux (*frontier models*). Le pipeline du présent projet était particulièrement challengeante considérant ses multiples expériences d'apprentissages à la fois supervisé, non-supervisé, et self-supervised. Nonobstant, d'un point de vue pratique, ce projet a considérablement consolidé mes compétences en ingénierie ML avancée, en particulier sur la gestion des tenseurs à haute dimension et l'optimisation des flux de données.